Type MS · Write or complete the respiration equations

Answer 1 of 9 · Question 5 part b · 3 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

5 (a) (i)	A = sternum; B = rib(cage); C = intercostal muscle;		ignore external/internal	3
(ii)	• lungs; • diaphragm; • nose/trachea/bronchi/bronchus/bronchioles/alveoli;		in any order mark only first three	3
(iii)	• diaphragm flattens; • ribs move up and out; • by contraction of intercostal muscles; • volume of thorax increases; • pressure in thorax/lungs decreases; • air forced/rushes/sucked/drawn, in (from outside);		reject moves down ignore external/internal ignore air flows in	6
(b)	• $C_6H_{12}O_6 + 6O_2$; • $6CO_2 + 6H_2O$; • energy/ATP;			3
				15 marks

Type MS · Write or complete the respiration equations

Answer 1 of 9 · Question 5 part b · 3 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Write or complete the respiration equations

Answer 2 of 9 · Question 1 part b · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

1	(a)	(i)	X cartilage; Y femur; Z knee cap;		1 1 1
		(ii)	**B** ligaments hold the leg bones together A is incorrect as ligaments do not attach muscles to bones C is incorrect as ligaments do not move leg bones D is incorrect as ligaments do not stop the leg bones rubbing together		1
		(iii)	hinge/synovial (joint)		1
	(b)		In the following order only oxygen; carbon dioxide;		 1 1
	(c)		biceps contracted/fatter; triceps relaxed/thinner; 		1 1

Total question 1 = 9 marks

Type MS · Write or complete the respiration equations

Answer 2 of 9 · Question 1 part b · 2 marks

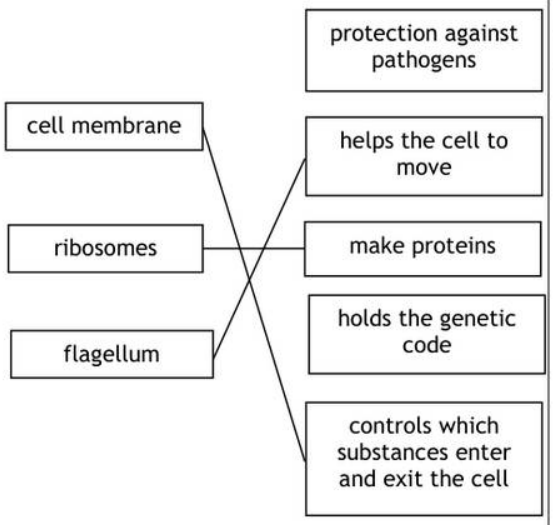
2019 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Write or complete the respiration equations

Answer 3 of 9 · Question 3 part b.ii · 2 marks

2020 JAN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

3 (a) (i)			Reject more than one line from each structure One mark for each correct line	3
	(ii)	Any two from: • (bacterial cell) has a cell wall (1) • (bacterial cell) does not have a nucleus (1) • (bacterial cell) has a plasmid (1)		2
(b)	(i)	**The only correct answer is B diffusion** *A is incorrect because it is down a diffusion gradient* *C is incorrect because it is not removal from the bacterium* *D is incorrect because osmosis only involves movement of water*		1
	(ii)	• glucose (1) • carbon dioxide (1)	Answers must be in the correct order Allow correct chemical formulae	2
	(iii)	Any two from: • to provide energy (1) • for movement (1) • for growth/reproduction/divide/mitosis (1) • for chemical reactions/producing organelles/named organelle (1)		2

Type MS · Write or complete the respiration equations

Answer 3 of 9 · Question 3 part b.ii · 2 marks

2020 JAN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Total for Question 3 = 10 marks

Type MS · Write or complete the respiration equations

Answer 4 of 9 · Question 2 part a · 4 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

number				
2	(a)	• oxygen; • glucose;	in any order	2
		• water; • carbon dioxide;	in any order	2
	(b) (i)	• A to B = 60 seconds; • 10 breaths during that period = 10 per minute;	Allow 60 anywhere in response	2
	(ii)	• 19.5 - 13.5 squares of movement = 6 squares; • $1dm^3$ = 4 squares; • 6/4 = $1.5dm^3$;		3
	(c)	• peaks/troughs higher/lower; • closer together;	Allow increase in amplitude Allow increase in frequency Answers must relate to peaks/waves/curves Ignore details relating to scientific details such as number or depth of breaths.	2

Total 11 marks

Type MS · Write or complete the respiration equations

Answer 4 of 9 · Question 2 part a · 4 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Write or complete the respiration equations

Answer 5 of 9 · Question 5 part a.i · 2 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

Question number					
5	(a)	(i)	$C_6H_{12}O_6$; $6H_2O$;		2
		(ii)	ATP breaks down/hydrolysed; to ADP and Pi;	Allow phosphate/P	2
	(b)	(i)	• suitable axis labels with units; • time on x axis; • suitable scale; • correct plots;		4
		(ii)	• suitable curve;		1
		(iii)	Five from; • mass of haemoglobin increases (with time/training at altitude); • increased number of red blood cells/more haemoglobin in the blood; • haemoglobin binds oxygen/red blood cells carry oxygen; • more oxygen transported around the body/in the blood; • to muscle (cells/tissue); • for aerobic respiration; • to release energy;		5

Total 14 marks

Type MS · Write or complete the respiration equations

Answer 5 of 9 · Question 5 part a.i · 2 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Write or complete the respiration equations

Answer 6 of 9 · Question 4 part a.i · 2 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

Number					
4	(a)	(i)	• glucose; • carbon dioxide and water;	Reject formulae	2
		(ii)	aerobic;		1
		(iii)	B; (mitochondrion)		1
		(iv)	ATP/adenosine triphosphate		1
	(b)	(i)	Any two from • more people/repeat tests: • different gender/diet/location/age; • carry out test over more years/longer time period; • calculate the mean/ average;		2
		(ii)	• 7250 - 6550;	ecf if wrong figures used in a subtraction calculation but correct answer 1 mark	
			• = 700kJ;		2
		(iii)	Any five from • metabolic rate increases in colder temperatures; • correct comparative reading from graph e.g. metabolic rate is higher in January compared to July; • (in colder temperatures) increased rate of respiration; • releases heat (energy)/more energy needed; • (energy needed) to maintain body temperature/reference to keeping body warm; • (in warmer weather) more heat lost (by radiation / convection/sweating);	ORA for mps 1, 3, 4, 5, 6 Allow marking point if correct figures given from graph	5
		(iv)	uptake of oxygen/carbon dioxide output;	Allow amount of glucose used	1

Total 15

Type MS · Write or complete the respiration equations

Answer 6 of 9 · Question 4 part a.i · 2 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Write or complete the respiration equations

Answer 7 of 9 · Question 4 part a.ii · 1 mark

2023 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number					
4	(a)	(i)	• glucose; • carbon dioxide and water;	Reject formulae	2
		(ii)	aerobic;		1
		(iii)	B; (mitochondrion)		1
		(iv)	ATP/adenosine triphosphate		1
	(b)	(i)	Any two from • more people/repeat tests: • different gender/diet/location/age; • carry out test over more years/longer time period; • calculate the mean/ average;		2
		(ii)	• 7250 - 6550;	ecf if wrong figures used in a subtraction calculation but correct answer 1 mark	
			• = 700kJ;		2
		(iii)	Any five from • metabolic rate increases in colder temperatures; • correct comparative reading from graph e.g. metabolic rate is higher in January compared to July; • (in colder temperatures) increased rate of respiration; • releases heat (energy)/more energy needed; • (energy needed) to maintain body temperature/reference to keeping body warm; • (in warmer weather) more heat lost (by radiation / convection/sweating);	ORA for mps 1, 3, 4, 5, 6 Allow marking point if correct figures given from graph	5
		(iv)	uptake of oxygen/carbon dioxide output;	Allow amount of glucose used	1

Total 15

Type MS · Write or complete the respiration equations

Answer 7 of 9 · Question 4 part a.ii · 1 mark

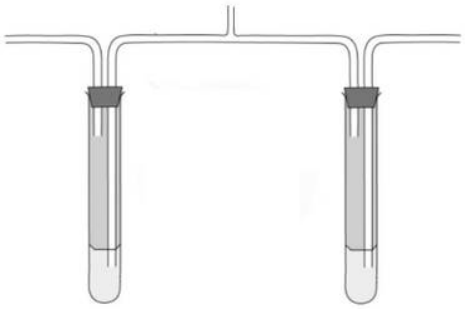
2023 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Write or complete the respiration equations

Answer 8 of 9 · Question 7 part a.i · 1 mark

2024 JUN · Paper 02 (regional R) · find part a.i in this mark scheme extract

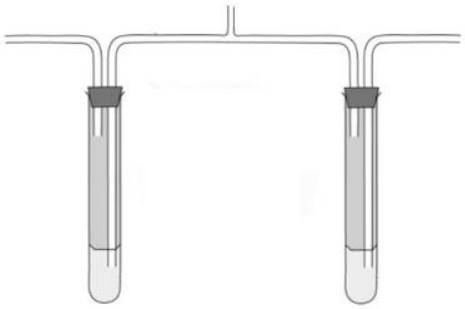
7 (a)	(i)	glucose ⟶ lactic acid/lactate;	ignore any ref to CO_2 and H_2O on right side of equation	(1)
	(ii)	$C_6H_{12}O_6 + 6O_2; \longrightarrow 6H_2O + 6CO_2;$		(2)
	(iii)	general diagram; tubing correct way round; 	labels not needed long short, long short short long, short long	(2)
	(iv)	• inhale and exhale/breathe through the mouthpiece; • at rest; • time how long it takes the limewater to go cloudy/milky; • repeat following exercise;		(4)
(b)		any five from • energy released from (breakdown of) glucose; • during (aerobic) respiration; • (energy used) to add phosphate group; • to convert ADP to form ATP; • energy is released from ATP breakdown of ATP releases energy; • breaking of phosphate bond/ATP hydrolysed/ATP breaks down into ADP + P(i); • to provide energy for muscle contraction;		(5)

Total for Question 7 = 14 marks

Type MS · Write or complete the respiration equations

Answer 9 of 9 · Question 7 part a.ii · 2 marks

2024 JUN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

7 (a)	(i)	glucose ⟶ lactic acid/lactate;	ignore any ref to CO_2 and H_2O on right side of equation	(1)
	(ii)	$C_6H_{12}O_6 + 6O_2; \longrightarrow 6H_2O + 6CO_2;$		(2)
	(iii)	general diagram; tubing correct way round; 	labels not needed long short, long short short long, short long	(2)
	(iv)	• inhale and exhale/breathe through the mouthpiece; • at rest; • time how long it takes the limewater to go cloudy/milky; • repeat following exercise;		(4)
(b)		any five from • energy released from (breakdown of) glucose; • during (aerobic) respiration; • (energy used) to add phosphate group; • to convert ADP to form ATP; • energy is released from ATP breakdown of ATP releases energy; • breaking of phosphate bond/ATP hydrolysed/ATP breaks down into ADP + P(i); • to provide energy for muscle contraction;		(5)

Total for Question 7 = 14 marks